

The Chain Rule



Basic chain rule applications

1 Differentiate each function using the chain rule:

a $f(x) = (4x - 1)^5$

b $f(x) = \sin(3x^2)$

c $f(x) = e^{5x}$

d $f(x) = \ln(2x + 7)$

2 Differentiate $f(x) = \sqrt{x^2 + 9}$.

The chain rule with trig and exponential functions

3 Differentiate $f(x) = \cos^3(x)$ (i.e. $[\cos x]^3$).

4 Differentiate $f(x) = e^{\sin x}$.

5 Differentiate $f(x) = \tan(e^x)$.

Nested (multiple) chain rule

6 Differentiate $f(x) = \sin(\sqrt{x^2 + 1})$, applying the chain rule twice.

7 Differentiate $f(x) = (\ln(3x))^4$.

Chain rule combined with product/quotient rules

8 Differentiate $f(x) = x^2 e^{3x}$.

9 Differentiate $f(x) = \frac{\sin(2x)}{x}$.

Chain rule with tables of values

10 Given $f(2) = 5$, $f'(2) = 3$, $g(2) = 7$, $g'(2) = -1$, find $\frac{d}{dx}[f(g(x))]$ at $x = 2$ if $g(2) = 7$ and $f'(7) = 4$ (note the composition uses f' evaluated at $g(2) = 7$, not at 2).

Chain rule with inverse trig and logarithms

11 Differentiate $f(x) = \sin^{-1}(2x)$.

12 Differentiate $f(x) = \ln(\sin x)$.

13 Differentiate $f(x) = \tan^{-1}(x^2)$.

Finding tangent lines using the chain rule

14 Find the equation of the tangent line to $f(x) = (2x - 1)^3$ at $x = 1$.

Visualizing a composite function and its derivative

15 The graph shows $f(x) = (x - 1)^2$ (an inner-outer composite of $u = x - 1$ and u^2), with its tangent line at $x = 2$.



a Use the chain rule to find $f'(x)$.

b Confirm $f'(2)$ matches the slope of the tangent line shown.

The chain rule in applied rate problems

- 16 The volume of a sphere is $V = \frac{4}{3}\pi r^3$, and the radius is growing according to $r(t) = 2t + 1$ (cm, seconds). Use the chain rule to find $\frac{dV}{dt}$ at $t = 2$.
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Chain rule with a table of composed values

- 17 Given $h(x) = f(g(x))$ where $g(1) = 3$, $g'(1) = 2$, $f(3) = 7$, $f'(3) = -4$, find $h'(1)$.
- 18 Differentiate $f(x) = \sqrt[3]{x^2 + 1}$ using the chain rule.
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Multiple choice — easy 15

- 19 Use the chain rule to differentiate $f(x) = (x - 1)^2$, then find $f'(1)$.
- a. 0
 - b. -2
 - c. 3
 - d. 5
- 20 Use the chain rule to differentiate $f(x) = (x - 2)^3$, then find $f'(0)$.
- a. 12
 - b. 4
 - c. 10
 - d. -24
- 21 Use the chain rule to differentiate $f(x) = (4x - 3)^2$, then find $f'(1)$.
- a. 4
 - b. 6
 - c. 2
 - d. 8

22 Use the chain rule to differentiate $f(x) = (2x - 2)^4$, then find $f'(1)$.

a. -2

b. 3

c. 0

d. 2

23 Use the chain rule to differentiate $f(x) = (4x - 1)^2$, then find $f'(1)$.

a. 24

b. 72

c. 12

d. 6

24 Use the chain rule to differentiate $f(x) = (3x + 1)^3$, then find $f'(0)$.

a. 14

b. 12

c. 3

d. 9

25 Use the chain rule to differentiate $f(x) = (3x)^4$, then find $f'(1)$.

a. 81

b. 324

c. 108

d. 972

- 26 Use the chain rule to differentiate $f(x) = (2x + 4)^4$, then find $f'(-2)$.
- a. 5
 - b. 3
 - c. -2
 - d. 0
- 27 Use the chain rule to differentiate $f(x) = (2x + 3)^4$, then find $f'(0)$.
- a. 108
 - b. 648
 - c. 216
 - d. 54
- 28 Use the chain rule to differentiate $f(x) = (x + 3)^4$, then find $f'(0)$.
- a. 27
 - b. 324
 - c. 110
 - d. 108
- 29 Use the chain rule to differentiate $f(x) = (x)^3$, then find $f'(1)$.
- a. 5
 - b. 1
 - c. 0
 - d. 3

- 30 Use the chain rule to differentiate $f(x) = (x - 1)^3$, then find $f'(0)$.
- a. 3
 - b. -3
 - c. 0
 - d. 1
- 31 Use the chain rule to differentiate $f(x) = (4x + 6)^2$, then find $f'(-2)$.
- a. -16
 - b. -8
 - c. -4
 - d. 32
- 32 Use the chain rule to differentiate $f(x) = (2x + 4)^2$, then find $f'(-2)$.
- a. -2
 - b. 0
 - c. 2
 - d. 5
- 33 Use the chain rule to differentiate $f(x) = (x + 1)^4$, then find $f'(1)$.
- a. 64
 - b. 8
 - c. 32
 - d. 35

Multiple choice — medium 15

- 34 Given $g(5) = 4$, $g'(5) = -2$, and $f'(4) = 5$, find $h'(5)$ if $h(x) = f(g(x))$.
- a. 5
 - b. -2
 - c. 3
 - d. -10
- 35 Given $g(7) = -5$, $g'(7) = -5$, and $f'(-5) = 2$, find $h'(7)$ if $h(x) = f(g(x))$.
- a. -5
 - b. -3
 - c. -10
 - d. 2
- 36 Given $g(6) = -5$, $g'(6) = -6$, and $f'(-5) = -4$, find $h'(6)$ if $h(x) = f(g(x))$.
- a. -6
 - b. 24
 - c. -4
 - d. -10
- 37 Given $g(5) = 0$, $g'(5) = -2$, and $f'(0) = -6$, find $h'(5)$ if $h(x) = f(g(x))$.
- a. 12
 - b. -8
 - c. -6
 - d. -2

38 Given $g(4) = -3$, $g'(4) = 6$, and $f'(-3) = 3$, find $h'(4)$ if $h(x) = f(g(x))$.

- a. 9
- b. 18
- c. 6
- d. 3

39 Given $g(7) = 3$, $g'(7) = 1$, and $f'(3) = 4$, find $h'(7)$ if $h(x) = f(g(x))$.

- a. 1
- b. 9
- c. 4
- d. 5

40 Given $g(3) = -3$, $g'(3) = 6$, and $f'(-3) = -4$, find $h'(3)$ if $h(x) = f(g(x))$.

- a. -24
- b. -4
- c. 6
- d. 2

41 Given $g(7) = -4$, $g'(7) = 3$, and $f'(-4) = 5$, find $h'(7)$ if $h(x) = f(g(x))$.

- a. 8
- b. 5
- c. 3
- d. 15

42 Given $g(8) = 6$, $g'(8) = -3$, and $f'(6) = -3$, find $h'(8)$ if $h(x) = f(g(x))$.

a. -6

b. 14

c. -3

d. 9

43 Given $g(3) = -5$, $g'(3) = 6$, and $f'(-5) = 2$, find $h'(3)$ if $h(x) = f(g(x))$.

a. 2

b. 6

c. 12

d. 8

44 Given $g(6) = -5$, $g'(6) = -3$, and $f'(-5) = 6$, find $h'(6)$ if $h(x) = f(g(x))$.

a. -18

b. -3

c. 3

d. 6

45 Given $g(2) = -5$, $g'(2) = 1$, and $f'(-5) = -5$, find $h'(2)$ if $h(x) = f(g(x))$.

a. -5

b. -4

c. -9

d. 1

- 46 Given $g(3) = 1$, $g'(3) = -1$, and $f'(1) = -4$, find $h'(3)$ if $h(x) = f(g(x))$.
- a. -5
 - b. 4
 - c. -1
 - d. -4
- 47 Given $g(3) = -6$, $g'(3) = -2$, and $f'(-6) = -3$, find $h'(3)$ if $h(x) = f(g(x))$.
- a. -5
 - b. -3
 - c. 6
 - d. -2
- 48 Given $g(6) = 6$, $g'(6) = 3$, and $f'(6) = -6$, find $h'(6)$ if $h(x) = f(g(x))$.
- a. 3
 - b. -3
 - c. -6
 - d. -18

Multiple choice — hard 15

- 49 Use the chain rule to differentiate $f(x) = \sqrt{x^2 + 16}$, then find $f'(-3)$.
- a. $-\frac{5}{3}$
 - b. $-\frac{3}{4}$
 - c. $-\frac{3}{5}$
 - d. $\frac{3}{5}$

50 Use the chain rule to differentiate $f(x) = \sqrt{x^2 + 144}$, then find $f'(5)$.

a. $\frac{13}{5}$

b. $-\frac{5}{13}$

c. $\frac{5}{13}$

d. $\frac{5}{12}$

51 Use the chain rule to differentiate $f(x) = \sqrt{x^2 + 144}$, then find $f'(-5)$.

a. $-\frac{5}{12}$

b. $-\frac{13}{5}$

c. $-\frac{5}{13}$

d. $\frac{5}{13}$

52 Use the chain rule to differentiate $f(x) = \sqrt{x^2 + 144}$, then find $f'(-9)$.

a. $-\frac{3}{5}$

b. $-\frac{5}{3}$

c. $\frac{3}{5}$

d. $-\frac{3}{4}$

53 Use the chain rule to differentiate $f(x) = \sqrt{x^2 + 16}$, then find $f'(3)$.

a. $-\frac{3}{5}$

b. $\frac{3}{4}$

c. $\frac{3}{5}$

d. $\frac{5}{3}$

- 54 Given $h(2) = -3$, $h'(2) = -3$, $g(-3) = -1$, $g'(-3) = -4$, and $f'(-1) = -3$, find $k'(2)$ if $k(x) = f(g(h(x)))$.
- a. -36
 - b. -10
 - c. 12
 - d. 9
- 55 Given $h(7) = 3$, $h'(7) = -2$, $g(3) = -5$, $g'(3) = -2$, and $f'(-5) = 2$, find $k'(7)$ if $k(x) = f(g(h(x)))$.
- a. 13
 - b. 8
 - c. -2
 - d. -4
- 56 Given $h(5) = 0$, $h'(5) = -5$, $g(0) = -5$, $g'(0) = 3$, and $f'(-5) = -4$, find $k'(5)$ if $k(x) = f(g(h(x)))$.
- a. 60
 - b. -12
 - c. -6
 - d. 20
- 57 Given $h(3) = 3$, $h'(3) = -3$, $g(3) = -4$, $g'(3) = -2$, and $f'(-4) = 5$, find $k'(3)$ if $k(x) = f(g(h(x)))$.
- a. -15
 - b. 0
 - c. -10
 - d. 30

- 58 Given $h(2) = 0$, $h'(2) = 2$, $g(0) = 1$, $g'(0) = -4$, and $f'(1) = -5$, find $k'(2)$ if $k(x) = f(g(h(x)))$.
- a. 40
 - b. -10
 - c. 20
 - d. -7
- 59 Which is the correct derivative of $F(x) = \sin(5x + 2)$?
- a. $5\cos(5x + 2)$
 - b. $-5\sin(5x + 2)$
 - c. $5\cos(x)$
 - d. $\cos(5x + 2)$
- 60 Which is the correct derivative of $F(x) = \sin(-2x - 1)$?
- a. $-2\cos(-2x - 1)$
 - b. $2\sin(-2x - 1)$
 - c. $-2\cos(x)$
 - d. $\cos(-2x - 1)$
- 61 Which is the correct derivative of $F(x) = \sin(-6x)$?
- a. $6\sin(-6x)$
 - b. $\cos(-6x)$
 - c. $-6\cos(x)$
 - d. $-6\cos(-6x)$

62 Which is the correct derivative of $F(x) = \sin(-6x + 4)$?

- a. $-6\cos(x)$
- b. $-6\cos(-6x + 4)$
- c. $\cos(-6x + 4)$
- d. $6\sin(-6x + 4)$

63 Which is the correct derivative of $F(x) = \sin(-5x - 2)$?

- a. $\cos(-5x - 2)$
- b. $-5\cos(x)$
- c. $-5\cos(-5x - 2)$
- d. $5\sin(-5x - 2)$